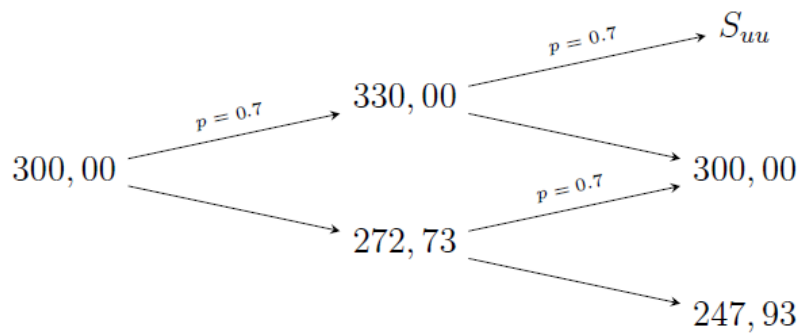


Exercise 1 (30%)

Mike Johnson has the most of his life had a big passion for stock, but he has of recently developed a growing interest for options. He is thinking of buying a very risky stock, based on his own analysis. Han can choose to buy the stock directly or indirectly, for example, through a call option. The stock has a price of 300 dkk today and can as of next year develop in the following ways.



Within the first 6 months, the price of the stock can either rise to 330 with a probability of $p = 0.7$ or fall to a price of 272.73 with a probability of $1 - p = 0.3$. The following six months can the stock then again rise or fall in price, in the way which is shown in the binomial tree above. The risk free rate with continuously compounding is $r = 4.879\%$ yearly.

Question 1) Calculate a) the smallest price of S_{uu} which ensures that the model is arbitrage free and b) the S_{uu} which implies that the risk neutral probabilities are constant through out the tree. Also assign the values of q and $q - 1$

Question 2) What is the price at time 0 on a European put option with strike $X = 295$? Use the value for S_{uu} which you found in 1.b to answer this question.

Question 3) Assume that the stock after six months has fallen to a share price of 272.73. Find the portfolio of the underlying asset and a risk-free asset, which replicates the put option the following six months.

Question 4) Use the put-call parity to calculate the price of the identical call option.

Question 5) If the parameters in the binomial tree are chosen according to the steps in Cox, Ross and Rubinstein, which value of the stock volatility σ have we then implicitly assumed? What shall the call option cost with this volatility according to the Black-Scholes equation? Why does this price diverge from the price you found in question 4?

Question 6) Assume that Mike can buy the call option for the price you found in question 4. Also assume – independent of the information above – that the price of the stock when option is expired after 1 year is 320. What investment has then reached the highest rate of return a) buy the stock or b) buy the option?

Exercise 2 (25%)

A bond market consists of three annuity bonds with yearly coupon rates. The following table indicates the other necessary information about the bonds.

Couponrate	Maturity	Share price
3%	1 <i>year</i>	98.880
4%	2 <i>year</i>	99.147
6%	3 <i>year</i>	101.758

Question 1) Write down the stream of payments for each of the three bonds, and explain whether the market is complete.

Question 2) Calculate the discount factors in the market. Decide if the market is arbitrage free.

Question 3) What is the spot rates in the market, s_1 , s_2 , s_3 ? What is the forward rate between year 2 and 3?

Question 4) Calculate the yield to maturity of the 3-year bond and its modified Macauley duration.

Question 5) Which share price should the 3-year bond approximately trade at if the spot rates increase by 1%-point?

Question 6) In the market there is also traded a 2-year bond which is of the type serial loan (the bond has equal principal payments). The bond has a coupon rate of 4%, and the price is 100 DKK. Is the bond priced correctly? If no, then propose an arbitrage strategy.

Exercise 3 (25%)

Mike Johnson have after further thought given up the idea of buying options. He is, however, still interested in buying the relevant stock, but he does not want to go all in. He has therefor found 2 other stocks, which future payments streams he is very positive about. The rate of returns of these stocks follow the distribution $r \sim N(\mu, \Sigma)$, where $r = [r_1, r_2, r_3]'$, r_i is the rate of return on stock "i", and it is also given that

$$\mu = \begin{bmatrix} 0,05 \\ 0,10 \\ 0,15 \end{bmatrix} \quad \Sigma = \begin{bmatrix} 0,010 & 0,006 & 0,015 \\ 0,006 & 0,040 & 0,042 \\ 0,015 & 0,042 & 0,090 \end{bmatrix}.$$

$$\Sigma^{-1} = \begin{bmatrix} 134.2105 & 6.5789 & -25.4386 \\ 6.5789 & 49.3421 & -24.1228 \\ -25.4386 & -24.1228 & 26.6082 \end{bmatrix}$$

$$\mu' \Sigma^{-1} \mu = 0.3882, \quad \mu' \Sigma^{-1} \iota = 5.5044, \quad \iota' \Sigma^{-1} \iota = 124.1959.$$

Question 1) Calculate the correlation between the rate of returns of asset 1 and 2. Explain, solely on this basis, if you can expect to diversify your risk buy investing in such a portfolio.

Mike has in the first instance chosen to combine a portfolio of the 3 stocks based on his own assessment of their potential. He has bought the portfolio $\omega = (0.3, 0.5, 0.2)'$

Question 2) Calculate the expected rate of return for this portfolio and the standard deviation. What is the possibility that this portfolio yields a negative rate of return?

Question 3) Given that Mike wants a rate of return of $\mu_p = 0.1$, what portfolio should he then invest in? What is the associated risk?

Question 4) Calculate the MVP and the expected rate of return and risk (deviation) of the MVP.

Question 5) Calculate the portfolio weights of the tangency portfolio under the condition that the risk-free asset is $r_f = 0.03$. Decide the rate of return and risk (deviation) of the tangency portfolio.

Question 6) Find the Capital Market Line and draw it in a (μ, σ) – diagram. Place the 3 assets and specify graphically which part of their risk is systematic and unsystematic.

Exercise 4 (20%)

Consider a one-period model for a financial market. There exist two assets, which both cost 1 at time 0. At time 1, the economy can be at three possible states. The assets value at time 1 is d_s^i , where the i is the asset and s is the state for $i = 1, 2$ and $s = 1, 2, 3$. The following is known:

$$\begin{aligned} d_1^1 &= 2, & d_2^1 &= 0, & d_3^1 &= 2 \\ d_1^2 &= 2, & d_2^2 &= 2, & d_3^2 &= 0 \end{aligned}$$

$\psi = (\psi_1, \psi_2, \psi_3)'$ denote the state prices.

Question 1) Write up the system of equations which the state prices shall uphold.

Question 2) Find a set of state prices. Determine if the financial market is arbitrage free and complete.

We now assume that a new asset is added to the economy. This asset costs 1 at time 0. This asset can at time 1 have the following values:

$$d_1^3 = 0, d_2^3 = 2, d_3^3 = 2$$

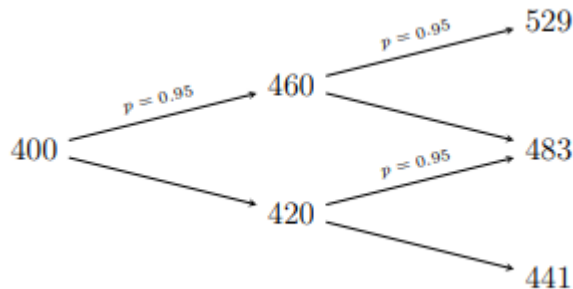
Question 3) Find a set of state prices in the expanded economy. Decide if the market is arbitrage free and complete.

Question 4) Calculate the risk neutral probabilities to end up in the 3 different states. Decide also what the risk-free rate should be in the economy, if such a risk-free asset exists.

Question 5) Assume that you can trade a forward contract on asset 1, with maturity date at time 1. The forward price is set to 1. What is the arbitrage-free price of the forward at time 0?

Exercise 1 (25%)

A stock has the price of 400 kr. today. The company have multiple products that are developing fast, which means there is a high probability for the price of the stock to increase further in the following years. To be more precise the stock can rise for the next two years with the probability p and fall with the probability $1-p$, in the following way.



Question 1: Assume the annual interest rate is 5% and compounded annually. Explain if this can cause any problems or if it won't affect the model.

For the rest of the assignment, you should assume that the annual risk-free rate is 8%.

Question 2: Compute the risk-neutral probabilities for the stock to rise or fall

Question 3: Determine the price of a 2-year European call and put option for the stock. Both options have strike $X = 485$

Question 4: Compute the replicating portfolio of the call option at time 0

Question 5: Prove that, $\Delta^{call} - \Delta^{put} = 1$ (Hint: Use the put-call parity)

You show your calculations from question 3 to a friend. If you haven't solved question 3, you can assume that the price of the call is 5 and the price of the put is 20. Your friend observes that the put is more expensive than the call, but that the two options give rise to almost identical payments at time 2, if the stock moves in the right direction from the perspective of the option. He feels that something is wrong with the prices. He thinks that the call should be more expensive than the put, because the stock almost certainly will increase in price over the next two years.

Question 6: Comment on your friend's statement. Is he right or wrong? And why?

Exercise 2 (25%)

Arne B. Thransen has been searching for new stocks for his private portfolio. He has found two stocks he will analyze further before including them in his existing portfolio.

The rate of return on these shares has the following characteristics:

$$\mu = \begin{pmatrix} 0.10 \\ 0.13 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} 0.0100 & 0.0045 \\ 0.0045 & 0.0225 \end{pmatrix}$$

with μ being the expected rate of return and Σ the covariance matrix.

Question 1: Arne buys the stake α and $1 - \alpha$ in the two stocks. Show that the rate of return of this strategy - as a function of α - has an expected return and variance (μ_p and σ_p^2) given by:

$$\begin{aligned} \mu_p &= 0.13 - 0.03\alpha \\ \sigma_p^2 &= 0.0225 - 0.0360\alpha + 0.0235\alpha^2 \end{aligned}$$

Question 2: State the value of α that minimizes the risk Arne is taking on by investing in the two stocks (the risk here is the standard deviation of the rate of return, σ_p). Compute μ_p and σ_p with this choice of α .

Arne reads a lot of entries on the internet about various shares. His eye has therefore caught sight of a third asset he will include in the analysis along the two other stocks. The rate of return on the third asset has an expected value of $\mu_3 = 0.16$ and a variance of $\sigma_3^2 = 0.0400$. In addition, it has a correlation of 0.5 with asset 1 and 0.7 with asset 2.

Question 3: Assume Arne invests in a portfolio of the three assets. His required rate of return is $\mu_p = 0.14$. Which portfolio should Arne buy? Compute the σ_p of this portfolio.

Question 4: Compute the portfolio weights in the tangency portfolio if the risk-free rate is $r = 0.07$. State the tangency portfolio's expected rate of return and risk.

Question 5: Assume that the CAPM assumptions are fulfilled. Compute β for the three assets and plot them in a diagram along with the security market line (SML). Explain the implications of an asset not being on the SML as well.

Exercise 3 (25%)

A bond market consists of three bonds with yearly payments. The bonds are described in the following table.

The spot rates for the market are given by $s_1 = 3.5\%$, $s_2 = 4.5\%$ and $s_3 = 6.0\%$ under the

Type	Coupon rate	Maturity
Coupon bonds	3%	3 years
Bond with equal principal payment each year	4%	2 years
Annuity	6%	3 years

assumption that there is annual accrual of interest.

Question 1: Write down the cashflows of the three bonds

Question 2: Compute the bonds theoretical prices and determine whether the market is arbitrage free and complete

Question 3: Assume, that you will enter a three-year interest rate swap, where the 1-year variable spot rates are swapped with a fixed interest rate, c . Compute the theoretical value of c .

You have gotten a bank to quote a forward on the three-year annuity bond, which you shall use to hedge your holdings. According to the contract, the bond can be traded immediately after the payment of time 2. The bank sets the forward price at 34.

Question 4: Compute the theoretical forward price. Explain if there is an arbitrage and how to realize it. Are there practical issues which can prevent you from exploiting the opportunity?

You decide to disregard the forward, but instead you use a futures contract to hedge your position. There is not being traded futures on the annuity bond, but instead on a generic bond which reflects the general development on the bond market. The variance of the return of the annuity bond is 9. The similar variance of the return of the future contract is 4. The correlation between these is 0.8

Question 5: What is the optimal hedge ratio, h ? If the future contract is used to hedge your holding, what is then the variance of the return of the total portfolio consisting of bonds and future contracts? How big of a percentage reduction, do you get because of the hedge?

Exercise 4 (25%)

Consider a model for an economy described by two stochastic factors: growth (F_1) and inflation (F_2). The factors measured as deviations (in %-points) from their expected value, so $E(F_1) = E(F_2) = 0$. There also exists three assets in the economy, which have the following relation between the factors and the rates of return on the assets:

$$\begin{pmatrix} r_1 \\ r_2 \\ r_3 \end{pmatrix} = \begin{pmatrix} 5 \\ 9 \\ 14 \end{pmatrix} + \begin{pmatrix} 1 & 2 \\ 3 & 3 \\ 4 & 2 \end{pmatrix} \begin{pmatrix} F_1 \\ F_2 \end{pmatrix},$$

where r_i is the rate of return on asset i , for $i = 1, 2, 3$.

Question 1: What is the expected rate of return on the three assets? If both growth and inflation are 1%-point higher than expected, then what is the realized rate of return on asset 1?

Question 2: Construct a portfolio that gives a risk-free return.

Question 3: Which values of $(\lambda_0, \lambda_1, \lambda_2)$ are required if APT is fulfilled? What is the risk-free rate in the economy if there exists a risk-free asset? Is this consistent with your answer to Question 2?

In the last part of the task, we study a fourth asset with a factor loading on the growth on 2, while the factor loading on the inflation is 2.5. This asset has an expected rate of return $\mu_4 = 6$.

Question 4: Give a well-founded explanation of whether it is generally against the arbitrage-free assumption, if a risky asset has a rate of return on level with the risk-free rate?

Question 5: Determine hereafter, if there in the specific cases of asset 4 is an arbitrage opportunity. If this is the case, you are to determine a strategy to take advantage of the opportunity.

Exercise 1 (25%)

A small bond market consists of three bonds that were originally issued many years ago, when interest rates were generally higher. The bonds all have annual terms and currently have a remaining maturity of 3 years. The following table provides a number of other information about the bonds for use in this assignment:

<i>Loan</i>	<i>Coupon</i>	<i>Effective interest</i>
Annuity	6%	4,7696
Serial bond (bond with equal principal payment each year)	8%	4,7061
Coupon bond	10%	5,7858

Question 1

Write down the payment series for the three bonds.

Question 2

Determine if the market is complete. Explain also, in your own words, what it means that a financial market is complete. Does your answer to the question change if the serial bond was instead an annuity bond?

Question 3

Calculate the price at which the bonds are traded in the market, and the associated discount factors. Determine, on that basis, whether the market is arbitrage-free.

Question 4

Calculate the modified duration of the three bonds.

Question 5

Based on a duration approximation, find the price range in which the bonds will trade, if the yield curve over the next year is expected to change by $\pm 1\%$ point at most

Question 6

Your longtime friend needs to place some liquid cash in the bonds next year. He says that “the best buy is the coupon bond as it gives the highest return of my money!” Discuss this statement.

Question 7

Assume that you can also trade a two-year zero-coupon bond at the price you found in question 3. Calculate the theoretical forward price of a 1-year forward on this bond. Interpret this price by explaining what aspect of the interest rate structure it represents?

Exercise 2 (20%)

Assume that at time t the price on a financial asset S is following a Geometric Brownian Motion:

$$dS_t = \mu S_t dt + \sigma S_t dW_t,$$

where W_t is a Brownian motion, while μ and $\sigma > 0$ are constants. Assume furthermore that $F_t = F(S_t, t) = \ln(S_t)$ is a function of S_t . Let r be the continuously compounded risk free rate.

Question 1

Which process does F_t follow?

Question 2

What is the distribution of $F_T = \ln(S_T)$ under the risk neutral probability measure?

Question 3

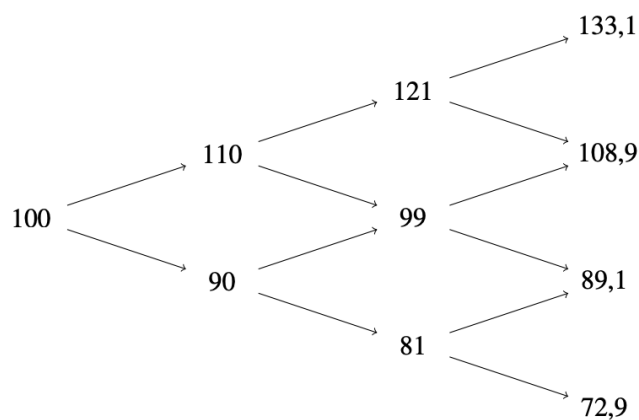
Assume instead that F is the payoff function on a European derivative asset with S as underlying and maturity T . Hence, a derivative which pays out the amount $F_T = \ln(S_T)$ at time T . What is the price of F at time t ?

Question 4

Verify that the price from question 3 fulfills the Black-Scholes equation. Explain why this is important.

Exercise 3 (25%)

Arne B. Thrasen has approached StorBank A/S to get some of his risk hedged. He wants to buy a European derivative with the stock S as the underlying asset and it expires in $T = 3$ years. The derivative pays $S_T - X_1$, if $S_T \geq X_1$ and $S_T - X_2$ if $S_T \leq X_2$, where $X_1 \geq X_2$. The stock is traded today at the price 100, and is expected to evolve as the figure shown beneath. It is known that there are no dividends in the period, and the *continuously* compounded risk-free interest is 2% per year.



Question 1

Find the risk neutral probabilities, q and $q - 1$.

Question 2

Compute the price of the derivative if $X_1 = X_2 = 100$ (this value is called X in the following). What type of derivative is it in this case?

Question 3

You propose that the derivative is instead issued with $X_1 = 105$ and $X_2 = 95$. If this is the occasion, what is the arbitrage free price of the derivative?

Question 4

What value does a European call option have with the strike $X = 100$?

Question 5

Specify, for the European call option, the composition up to time 3 of the replicating portfolio in the underlying active and a bank loan, if the price is 99 at time 2

Mr. Thrasen is also interested in option theory. He has therefore given you the following assignment as a kind of test, as he evaluate that it must be a small matter for a sharp mathematical economist as you:

Question 6

Let $C_t = S_t \Phi(d_1) - X e^{-r(T-t)} \Phi(d_2)$ be the price of the European call option in Black-Scholes model. Proof, by using the Put-Call parity (and nothing else), that the price of the corresponding European put option is $P_t = X e^{-r(T-t)} \Phi(-d_2) - S_t \Phi(-d_1)$ [Hint: $\Phi(x) + \Phi(-x) = 1$ for the standard normal cumulative distribution function].

Exercise 4 (30%)

Consider an economy consisting of three assets where the following information is available:

$$\Sigma = \begin{bmatrix} 0,20 & 0,05 & 0,15 \\ 0,05 & 0,10 & 0,05 \\ 0,15 & 0,05 & 0,15 \end{bmatrix}, \quad \Sigma^{-1} = \begin{bmatrix} 20 & 0 & -20 \\ 0 & 12 & -4 \\ -20 & -4 & 28 \end{bmatrix}, \quad \mu = \begin{bmatrix} 0,20 \\ 0,12 \\ 0,16 \end{bmatrix},$$

where μ is the assets expected rate of return, Σ is the covariance matrix of the assets. Furthermore, is it given that:

$$\iota' \Sigma^{-1} \iota = 12, \quad \mu' \Sigma^{-1} \iota = 1,6, \quad \mu' \Sigma^{-1} \mu = 0,2560, \text{ hvor } \iota = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}'.$$

The risk free rate is given to $r_f = 2\%$

Question 1

Calculate the portfolio weight, the expected rate of return and the standard deviation of the minimum variance portfolio. (MVP)

Question 2

Find the tangency portfolio and calculate the expected rate of return and the standard deviation.

Question 3

Is it possible to achieve an expected rate of return of 0.20 if the maximum risk (standard deviation) you want to accept is 0.40? If yes, then state the weights for the portfolio that achieves this return. If no, then state the highest return that can be achieved given the risk limitation.

You assume that there exists more than one risk factor for which the investors require compensation. You can estimate the following factor mode:

$$\begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix} = \begin{bmatrix} 0,20 \\ 0,12 \\ 0,16 \end{bmatrix} + \begin{bmatrix} 2 & 1 \\ 0,5 & 1 \\ 0,7 & 1,4 \end{bmatrix} \begin{bmatrix} F_1 \\ F_2 \end{bmatrix},$$

Where F_1 and F_2 are factors with mean value 0. It is also stated that:

$$\begin{bmatrix} 1 & 2 & 1 \\ 1 & 0,5 & 1 \\ 1 & 0,7 & 1,4 \end{bmatrix}^{-1} = \begin{bmatrix} 0 & 3,5 & -2,5 \\ \frac{2}{3} & \frac{-2}{3} & 0 \\ \frac{-1}{3} & \frac{-13}{6} & 2,5 \end{bmatrix}.$$

Question 4

Use the three assets 1, 2 and 3 to compute to pure factor portfolios P_1 and P_2 . That means you must calculate two sets of weights $(\omega_1, \omega_2, \omega_3)$, so that the associated portfolio satisfies the factor equation:

$$r_{P_1} = \mu_{P_1} + F_1,$$

$$r_{P_2} = \mu_{P_2} + F_2.$$

Question 5

State the expected rate of return and standard deviation of the two portfolios. What are the risk premiums, λ_1 and λ_2 , in this model?

Assume that there is introduced a fourth asset, which have the factor equation

$$r_4 = 0,18 + 1,5F_1 + F_2$$

Question 6

Given your knowledge of the risk-free interest rate and the risk premiums you calculated in the previous question, explain whether the expected rate of return on asset 4 is consistent with the absence of arbitrage?

Question 7

Compound a portfolio consisting of assets 1,2 and 3 that replicates asset 4 and explain how you can use the replicating portfolio to exploit a possible arbitrage opportunity.

Exercise 1

Consider a financial market, where the forward rate structure can be described as:

$$f_{0,1} = 0.05 \quad , \quad f_{1,2} = 0.04 \quad , \quad f_{2,3} = 0.06$$

1. Find the prices for the 1-, 2- and 3- years zero coupon bonds
2. Compute the arbitrage free price for a bond with the coupon rate of 3% and a maturity of 3 years.
3. Assume, that a new serial bond (same annual principal payment) and with a maturity of 3 years is bought for 99. What should the coupon rate be so that there is no arbitrage in the market?

Two new bonds with maturities of 5 years are introduced. One of them has a coupon rate of 3% and is priced at 95.47. The other bond has a coupon rate of 5% and is priced at 100.94.

4. Assume that both bonds are annuities. Contemplate whether the market is complete. Will the answer change if one of them is instead a serial bond?

Assume now that that the bond with the coupon rate of 5% is a serial bond and therefore has a fixed principal payment per year, and that the bond with the coupon rate of 3% is an annuity.

5. Write down the cashflows of the two bonds.
6. Compute the discount factors for year 4 and year 5.

Exercise 2

Consider a market, where 3 assets exist. It is known that there is a 25% chance that in the next period, the value of stocks 1, 2, and 3 will be 80, 90, and 25, respectively. Moreover, there is 50% chance that the values of the stocks 1, 2, and 3 will be 90, 64, and 35, respectively. Finally, there is a chance that the values of the stocks 1, 2, and 3 will be 35, 55, and 96, respectively. In addition, the following information is available:

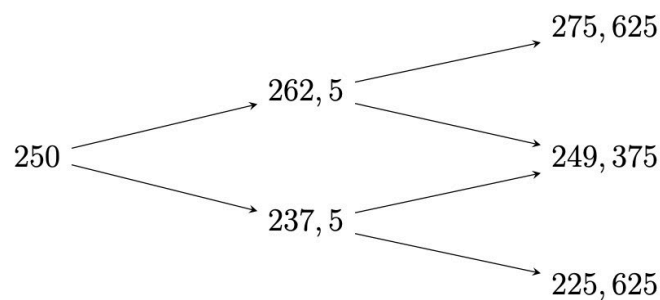
$$\begin{bmatrix} P_1 \\ P_2 \\ P_3 \end{bmatrix} = \begin{bmatrix} 54.25 \\ 58.75 \\ 56.95 \end{bmatrix}, \quad D^{-1} = \begin{bmatrix} -0.0160981 & 0.0277205 & -0.0059142 \\ 0.0282929 & -0.0259654 & 0.0020986 \\ -0.0103404 & 0.0047695 & 0.0113706 \end{bmatrix}$$

1. Find the state prices of the market. Is the market arbitrage free?
2. Conclude whether there are assets in the market, which have a return that cannot be replicated using the assets 1-3.
3. Compute the risk-free rate.
4. Find the expected returns of the assets. Discuss if it is worth investing in asset 3.
5. Consider a futures contract on asset 1. What is the futures price?

6. Consider a call option on asset 3 with strike $X=30$. Construct a portfolio that replicates the payoff of the call option. What is the fair price of the call option?

Exercise 3

Arne B. Thransen is again looking at the derivatives market. He is interested in a stock that is priced at 250 dkk. In the next couple of years, the stock is expected to have the following development:



1. Compute the interval, which the continuously compounded risk-free rate should be in, in the absence of arbitrage. Moreover, contemplate if it is consistent with no-arbitrage if the risk-free rate is negative.

For the rest of the exercise it can be assumed that $r_f = 2\%$ per year and compounded continuously.

2. Compute the risk-neutral probabilities.
3. Compute the price of a 2 year at-the-money European call option.
4. Compute the replicating portfolio of the call option up to period 1.
5. Assume that the otherwise identical put option costs 2.25. Show that this creates an opportunity for arbitrage. Explain the transactions that you would need to do to exploit the arbitrage
6. Let $\Gamma^{call} = \frac{\partial^2 C_t}{\partial S_t^2}$ be the gamma of the option. Show that $\Gamma^{call} = \Gamma^{put}$. (Hint: make use of the put-call parity)

Exercise 4

Consider two stocks A and B, that have returns with the following properties:

$$\mu_A = 0.1, \mu_B = 0.2, \sigma_A^2 = 0.16, \sigma_B^2 = 0.09,$$

where μ represents the expected rate of return, and σ^2 is the corresponding variance. It is assumed, that the returns of the stocks are uncorrelated.

1. Assume that you buy α and $(1 - \alpha)$ of the two stocks. Show, that the return of this strategy as a function of α has an expected return and variance given by:

$$\begin{aligned}\mu_p &= 0.2 - 0.1\alpha \\ \sigma_p^2 &= 0.09 - 0.18\alpha + 0.25\alpha^2\end{aligned}$$

2. Let $\alpha^* = \arg \min_{\alpha} \sigma_p^2$, be the value of α that minimizes the risk attributed to investing in the two stocks. Show that α^* is given by:

$$\alpha^* = \frac{\sigma_b^2}{\sigma_A^2 + \sigma_B^2}$$

3. Compute μ_p and σ_p , when the portfolio weight is α^* .
4. Assume that the assumptions of CAPM is fulfilled in the market. Explain if it is consistent with the CAPM theory, that a stock has a higher expected return, but lesser risk, than another stock. In particular, that $\mu_a < \mu_B$ and $\sigma_A^2 > \sigma_B^2$.
5. Assume that the risk-free rate is 5%. Find the market portfolio.
6. Compute the β for the stocks.

Exercise 1 (25%)

Your parents have sold their home and are left with a considerable amount of free funds which must be invested. You have agreed to draw up proposals for investing the money in the following three asset classes: government bonds (gov), mortgage bonds (reals) and Danish stocks (stocks). On the basis of the latest forecast for the Danish economy, made by the Danish central bank, you have drawn up four possible scenarios for asset returns over the coming year:

Scenario	Probability	Rate of return		
		Gov	Reals	Stocks
1	0.10	-0.05	0.15	-0.20
2	0.30	0.00	0.06	-0.05
3	0.40	0.06	-0.05	0.15
4	0.20	0.08	0.15	0.50

Question 1:

Calculate the assets' expected rate of return and risk (standard deviation).

Question 2:

Write down the covariance- and correlation matrix. If, based on this analysis, you had to choose exactly two assets to invest in, which would you choose? Justify your answer (no further calculations are required for the last part of this question).

Question 3:

Suppose you actually choose to invest in real estate and stocks. You buy 25% real estate and 75% stocks. What is the expected rate of return and risk of this portfolio?

Question 4:

Your parents want the least possible risk, as they will be retiring soon. Which portfolio of real estate and stocks fulfills this purpose?

In the following, we consider government bonds as risk-free assets. That means we will now assume that they provide the expected rate of return you calculated in question 1, in all four scenarios, instead of those indicated in the table

Question 5:

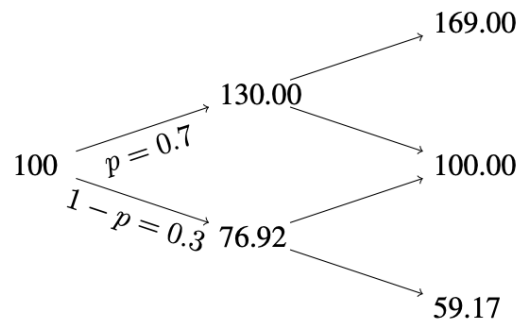
What are the weights in the tangent portfolio between reals and stocks? What Sharpe ratio does it have?

Question 6:

Explain everything you know about the term "Sharpe ratio" to your parents. How and why is this key term used to evaluate and compare different investments?

Exercise 2 (25%)

Arne B. Thrassen misses excitement in his everyday life. He has therefore intercepted a very volatile asset which has several relatively liquid options that are trading with the asset as the underlying asset. Arne is interested in trading the options, but he misses a double check on the prices that the banks set. He therefore needs your help. For information, the asset costs DKK 100 and it can develop in the following way over the next two years:



p denotes the probability of an increase in the asset, and is kept constant throughout the tree. In addition, it can be stated that the risk-free interest rate under continuous compounding is $r = \ln(1.035)$ per year. The asset does not pay dividends.

Question 1:

Show that the market is arbitrage-free and complete. Calculate the risk-neutral probabilities, q and $1 - q$.

Question 2:

What should a European call option with strike $X = 95$ and expiration at time 2 cost in this market? You also need to price the otherwise identical put option.

Question 3:

What should the American put option cost?

Question 4:

Compute the time-0 replicating portfolio of the American put option up to time 1.

Question 5:

Now assume that the asset trades continuously in the market. The price follows the process:

$dS_t = \mu S_t dt + \sigma S_t dW_t$. Explain whether it is compatible with the absence of arbitrage, if one of the derivative assets has the price: $F(S_t, t) = e^{S_t}$.

Exercise 3 (25%)

In this exercise, we consider an economy where the returns on financial assets follow a 2-factor model. The two factors are stochastically independent with $\tilde{F}_j \sim N(1, 1)$, $j = 1$ and 2 . Let r_i denote the rate of return on asset i . We initially look at a subset of the traded assets, which have the following structure:

$$\begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix} = \begin{bmatrix} 5 & 2 & 5 \\ 8 & 3 & 2 \\ 3 & 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ \tilde{F}_1 \\ \tilde{F}_2 \end{bmatrix},$$

It can also be shown that,

$$\begin{bmatrix} 1 & 2 & 5 \\ 1 & 3 & 2 \\ 1 & 1 & 3 \end{bmatrix}^{-1} = \begin{bmatrix} -1.4 & 0.2 & 2.2 \\ 0.2 & 0.4 & -0.6 \\ 0.4 & -0.2 & -0.2 \end{bmatrix}.$$

Question 1:

What is the expected rate of return of the assets?

Question 2:

What probability distribution does the rate of return of the assets follow?

Question 3:

What is the first asset's correlation with the following two factors in the economy ρ_{r_1, F_1} and ρ_{r_1, F_2} ?

Question 4:

Rewrite the model so that the factors are given as deviations from their expected value (Hint: you can call the standardized factors F_j , $j = 1$ and 2 . Note that $F_j = \tilde{F}_j - 1$).

Question 5:

Which values of $(\lambda_0, \lambda_1, \lambda_2)$ are consistent with APT? What is the risk-free interest rate in the society?

Consider a fourth asset that, at its current market price, has a rate of return of 10, if both the normalized factors F_1 and F_2 are zero. The asset has a factor loading on both factors, which is 2.

Question 6:

Explain whether the asset, at its current rate, is correctly priced. If not, you have to specifically state if the rate is too high or low compared to the equilibrium price.

Exercise 4 (25%)

Consider a bond market where three bonds with yearly payments are traded. The following table contains a number of necessary information about the bonds:

Bonds	Coupon rate	Maturity	Price
Coupon bond	5%	3 years	101.900
Serial loan (equal principal payments per year)	5%	2 years	99.850
Annuity bond	5%	3 years	102.084

Question 1:

Write down the payment rows on the bonds.

Question 2:

Calculate the discount factors. Explain whether the market is arbitrage-free and complete.

Question 3:

Consider a forward contract on the coupon bond, where you trade the bond for takeover shortly after the coupon payment at time 2. What is the arbitrage-free forward price?

Assume that there is a brand new 2-year annuity loan on the market. Similar to the 3-year annuity loan, the bond pays a coupon rate of 5% and trades at a price of 102.084.

Question 4:

Explain whether this gives rise to arbitrage. If yes, then calculate the replicating portfolio (of the three other bonds). If not, then explain why annuity loans with the same coupon rate, but different maturity should theoretically cost the same.

Question 5:

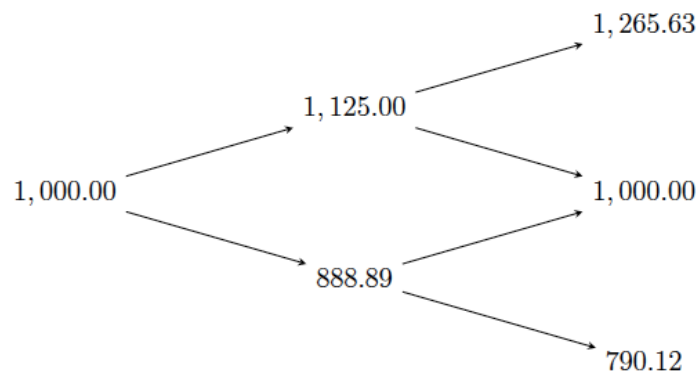
Calculate the modified duration of the coupon bond.

Question 6:

Based on a duration approximation, find the equilibrium price of the coupon bond if the interest rate structure increases by 1% point.

Assignment 1 (25%)

The Danish stock market, measured by the OMX C20 index, has had a large increase the last five year and has almost been doubled in value from a price of around 500 to a current level around 1,000. Henrik V. thinks that it is starting to look like a larger correction in the stock prices is coming up. Therefore, he plans to trade financial derivatives, where he earns money, if his predictions hold true. Henrik has found an analysis, which based on historical prices, has projected the possible price development of the stock market the next two years:



It is given in the analysis that $r_f = 2.5\%$ p. per year, with yearly compounding.

Q1: Show that the development in the Danish stock market is arbitrage-free. Calculate the risk neutral probabilities for an “up” and “down” movement in the market q and $1 - q$.

Q2: Price a two-year European put option with strike price of $X = 900$.

Q3: Calculate the replicating portfolio of the put option from time 0 to time 1.

Q4: Find the futures price at time 0.

Q5: Which payments does a futures contract make at time 1 and 2?

Q6: Consider a 1-year option, with payoff of 0, if the stock market increases in price, and with payoff of 1, if the stock market decreases in price (a binary put option). Show that the price of the binary put option reflects the risk-free probability of a “down” movement (up till discounting).

Assignment 2 (25%)

On a bond market, there are three bonds trading; a 1-year coupon bond, a 2-year serial bond (a bond with equal principal payments per period), a 3-year annuity bond. All of these are with yearly terms and a coupon rate of 10%.

Q1: Write up the payment streams for the three bonds. Explain if the market is complete.

The spotrate structure, that is computed from the total amount of traded bonds, is estimated amongst the marketparticipants to the following: $s_1 = 11.1111\%$, $s_2 = 11.8034\%$ and $s_3 = 12.6248\%$.

Q2: Compute the discount factors in the market. Is the market arbitrage-free? Also specify the equilibriumprices for the three bonds, that are consistent with the absence of arbitrage.

Q3: Compute the 1-year forward rates between time 1 and 2 and time 2 and 3.

Q4: Consider a 2-year forward on the annuity bond, where the bond is traded with takeover directly after the dividend at time 2 is paid. What is the arbitrage-free forward price?

Q5: Consider an investor, who is long in a 3-year bond of the type F1 (i.e., with “variable” interest rate, that is determined one year at a time). He wishes to swap the floating interest payments for a fixed rate during the entire term of the loan. What is the interest rate that he must pay after the swap?

Q6: Assume, that one of the other bonds (a 3-year coupon bond with coupon rate 10%) is traded in the market at price 97. Show that this is a possibility for arbitrage. Specify which combination of the three bonds above, you should buy, if you want to replicate the payment profile of the 3-year coupon bond.

Assignment 3 (25%)

The earlier mentioned analysis of the Danish stock market (from Assignment 1) also includes information about the individual subsidiary companies that constitute the index. Henrik V. has identified three stocks that he finds interesting, despite his view of the stock market being overvalued in general. Directly from the report the three stocks have expected returns and a covariance matrix given as:

$$\mu = \begin{bmatrix} 0.15 \\ 0.20 \\ 0.25 \end{bmatrix} \quad \text{and} \quad \Sigma = \begin{bmatrix} 0.0400 & 0.0180 & 0.0400 \\ 0.0180 & 0.0900 & 0.0840 \\ 0.0400 & 0.0840 & 0.1600 \end{bmatrix}$$

It is disclosed that

$$\Sigma^{-1} = \begin{bmatrix} 33.5526 & 2.1930 & -9.3595 \\ 2.1930 & 21.9298 & -12.0614 \\ -9.3595 & -12.0614 & 14.9671 \end{bmatrix}$$

Q1: Compute the expected return and standard deviation of a portfolio where the portfolio weights are proportional to the reciprocal of the volatility, $w_i = \frac{s_i}{\sum_{j=1}^n s_j}$, where $s_i = 1/\sigma_i$, $i = 1, \dots, n$.

Q2: The portfolio from Q1 is a global minimum variance portfolio in a special case. Explain which case (remember, that $w_{MVP} = \frac{1}{a} \Sigma^{-1} \mathbb{1}$, where $\mathbb{1} = (1, 1, \dots, 1)^T$ and $a = \mathbb{1}^T \Sigma^{-1} \mathbb{1}$).

Q3: Calculate now the weights in the actual minimum variance portfolio, as well as the expected return and the standard deviation. Compare your answer with Q1.

The return on the risk-free asset is, as in Q1, determined to be 2.5% per year.

Q4: Find the portfolio of stocks with the highest Sharpe ratio of all portfolios consisting of risky assets. Specify the expected return and the standard deviation.

Q5: Assume CAPM is applicable in the market. Find the expected return on an efficient asset, which has $\beta = 1.5$. What is the associated risk?

Q6: If the asset from Q5 has an expected time 1 value of $E(P_1) = 120$, what is the forward price on buying the asset on term at time 1? (If you have not answered the previous question, you must make appropriate assumptions.

Assignment 4

Assume that in the next period there are three main outcomes for the growth of the market, these we label as high, medium and low. We have three assets that have known value at time 0, and gives a growth dependent return at time 1. This is illustrated in the system below

$$P = \begin{bmatrix} 11.5 \\ 6.0 \\ 2.0 \end{bmatrix}, \quad D = \begin{bmatrix} 0 & 10 & 20 \\ 10 & 20 & 0 \\ 20 & 0 & 10 \end{bmatrix}$$

Q1: decide if the market is complete and arbitrage free.

Assume that after one second the market prices have changed, the Matrix D has not changed though. The new prices for the assets, which is to be used in the next assignments, is now $P = [11.5, 9.5, 9]'$

Q2: decide again if the market is complete and arbitrage free.

Q3: calculate the risk-free rate.

Q4: find the risk neutral probabilities in the market.

Q5: Consider a call option with strike $X = 10$ on asset 1. What is the price of the option?

Q6: Make a portfolio with the three assets that replicates the payments of the option.

Investment and Finance

June 2019

Opgave 1 (30%)

In this assignment, a standard binomial model is considered for the price of a dividend-free share, S_t , which over the next four quarters may develop as follows:

				73.21
			66.55	
		60.50		63.22
	55.00		57.48	
50.00		52.25		54.60
	47.50		49.64	
		45.13		47.16
			42.87	
				40.73
$t = 0.00$	$t = 0.25$	$t = 0.50$	$t = 0.75$	$t = 1.00$

In the entire tree, it is equally likely whether the stock rises or falls ($p = 1 - p = 0.5$). It can also be stated that the risk-free rate is zero. On this background, you must answer the following questions:

1. Show that the model is arbitrage-free and complete.
2. The rate of return $R_1 = \frac{S_1}{S_0} - 1$ is a stochastic variable as of time 0. Calculate the expected value of R_1 under the objective probability measure (P-measure), $E^P(R_1)$.
3. Write down the risk-neutral probabilities, q and $1 - q$.
4. Also calculate the expected value of R_1 under the Q measure, $E^Q(R_1)$. Interpret your result.
5. What is the time 0 price of a European call option with exercise price $X = 60$.

6. You have sold the above call option and will continuously hedge your risk. At time $t = 0.75$ the share has risen to rate 57.48. What does the replicating portfolio consist of at that time?
7. The above binomial grid is constructed on the basis of two unknown parameters μ and σ , that fulfills $u = e^{\mu\Delta t + \sigma\sqrt{\Delta t}}$ and $d = e^{\mu\Delta t - \sigma\sqrt{\Delta t}}$. Calculate the value of μ and σ .

Opgave 2 (20%)

A small financial market consists of three assets, which at time 0, costs $P_1 = 8.35, P_2 = 10.75, P_3 = 13.45$. At time 1, each asset can take on three different values given by the following matrix:

$$D = \begin{bmatrix} | & | & | \\ d^1 & d^2 & d^3 \\ | & | & | \end{bmatrix} = \begin{bmatrix} 2 & 3 & 5 \\ 7 & 11 & 13 \\ 17 & 19 & 23 \end{bmatrix}$$

The probabilities for the three different states are $\pi_1 = 0.25, \pi_2 = 0.50, \pi_3 = 0.25$

1. Calculate the state prices $\psi = (\psi_1, \psi_2, \psi_3)'$.
2. Explain, whether the market is arbitrage free and complete.
3. Calculate the risk-free rate, r , and the risk-neutral probabilities, $q_i, i = 1, 2, 3$

A European binary put option is a derivative that pays 1, if the price of the underlying asset j is less than or equal to the strike price, and otherwise it pays 0. Mathematically, it can be defined as:

$$d_i^{\text{binær}} = \begin{cases} 1 & \text{for } d_i^j \leq X, \\ 0 & \text{ellers.} \end{cases}$$

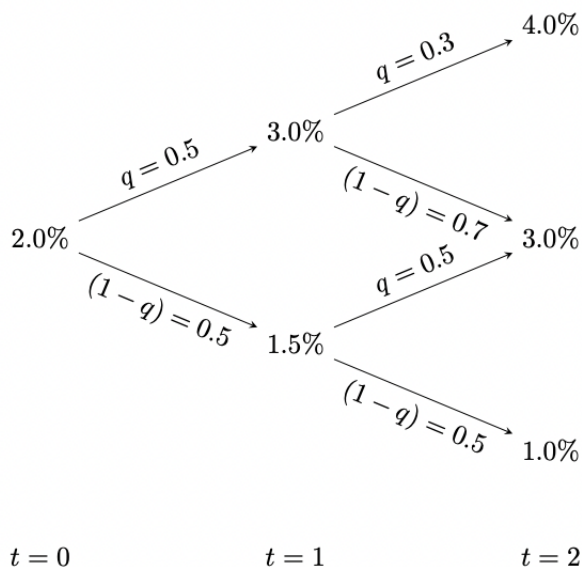
4. Price a European binary option on asset 3 with strike $X = 10$.

Consider the fraction $v_i = \frac{q_i}{\pi_i}, i = 1, 2, 3$. v_i denotes the ratio between the objective (P-measure) and the risk-neutral (Q-measure) probabilities. At time 0 this quantity can be considered as a stochastic variable v , that can take on 3 different values, depending on the realized states at time 1.

5. Calculate the expected value of v under the P-measure, $E^P(v)$

Exercise 3 (25%)

On a market, the stochastic interest rate over the next 2 periods is expected to follow the binomial model described in the figure below. The risk-neutral probabilities are indicated in the figure and the interest is compounded periodically.



1. Calculate the time 0 prices of the zero-coupon bonds maturing at times 1, 2 and 3. Show that the interest rate structure is given by $s_{0,1} = 2.00\%$, $s_{0,2} = 2.12\%$ and $s_{0,3} = 2.29\%$
2. In the market there is a 3-year bond (of the serial loan type which has equal principal payments each period) with a principal of 1000 and a price of 1015. Determine the coupon rate on the bond that is consistent with the absence of arbitrage.
3. There is a European call option with expiration at time 2 that has the bond as the underlying asset. Strike is $K = 332$. Calculate the price of the option.
4. Use the put-call parity to find the value of the corresponding European put-option.
5. Calculate the price of a 2-year futures contract on the bond (with delivery after the payment at time 2).

6. In the market, the corresponding 2-year forward contract is traded, so that forward price = futures price. Explain whether this gives rise to an arbitrage. If yes, outline how the opportunity can be exploited.

Opgave 4 (25%)

Spotlight is the Nordics' leading partner platform for smaller companies (often with strong growth and low profitability) that need access to capital via the stock market. As a private investor, you have searched the price list and located three shares that you plan to invest your saved funds in. There is the following data basis:

$$\mu = \begin{bmatrix} 0.06 \\ 0.08 \\ 0.10 \end{bmatrix} \quad \Sigma = \begin{bmatrix} 0.0144 & 0.0180 & 0.0234 \\ 0.0180 & 0.0400 & 0.0330 \\ 0.0234 & 0.0330 & 0.0900 \end{bmatrix}$$

The risk free rate is 2%

1. If, for various reasons, you have to invest all your funds in one of the three assets, which asset would you choose? You must justify your answer
2. Suppose you buy a portfolio where the portfolio weights are proportional to the reciprocal of the risk of the assets (standard deviation). Calculate the expected rate of return and risk of the investment
3. Calculate the portfolio weights of the global minimum variance portfolio of the risky assets (in this question you can not invest in the risk-free asset). State its expected rate of return and risk
4. Calculate the portfolio weights of the tangency portfolio and also state its expected rate of return and risk. What is the market price of risk?
5. Assume that the CAPM is satisfied in the market. Calculate the β 's of the three assets
6. Find the beta of the portfolio from question 2

Exercise 1

Assume, that the price of an asset is defined by $S_t = W_t$, where W_t is a standard Brownian motion (as always $W_0 = 0$). Consider the stochastic process $X_t = S_t^2 = W_t^2$

1. Use Itô's lemma to make an expression for dX_t
2. Find the mean and the variance of X_t (remember: $W_t \sim N(0, t)$)
3. Show that, $P(X_t \leq x) = 1 - 2\Phi\left(-\sqrt{\frac{x}{t}}\right)$, (for $t > 0$ and $x > 0$, where $\Phi(\cdot)$ is the cumulative distribution function for a standard normally distributed stochastic variable.
4. Explain, whether it is consistent with absence of arbitrage, if X_t is the price of a derivative asset.

Exercise 2

In this exercise, we consider a binomial model for the stochastic development in the short rate under the neutral risk probability measure. The following tree indicates the 1-year spot rate on different times and states

	$r_{1,2}(1) = 6\%$	$r_{2,3}(2) = 9\%$
$r_{0,1}(0) = 4\%$		$r_{2,3}(1) = 7\%$
	$r_{1,2}(0) = 3\%$	$r_{2,3}(0) = 2\%$
$t = 0$	$t = 1$	$t = 2$

Here $r_{t,t+1}(s)$ is the rate between time t and $t+1$ in state s , and with yearly compounding. The current 1-year rate is therefore 4%. Moreover, you must assume, that the risk neutral probability for an upward or downward movement of the short rate is $q = 1/2$ in the whole tree. Under this assumption, you need to answer the following:

1. Calculate the prices on zero coupon bonds in the market.
2. Compute the initial spot rate term structure.
3. Assume, that in the market, a 3-year inconvertible annuity bond with coupon of 4% is traded. What is the arbitrage free price on the bond (per 100 kr. nominal)?
4. What is the price on the corresponding convertible bond? Determine the explicit value of the convertible option, and explain at what time and state it is converted.
5. Calculate the state prices for the three possible states at time 2.
6. Consider a 2-year European put option with strike $X = 34$ on the inconvertible annuity loan. This option can be exercised after the payment at time 2. Calculate the equilibrium price of the option.

Exercise 3

Consider a market, where three assets, A, B and C, are traded. You have reviewed the future prospects for the different assets and on that background, you have set up a forecast for expected returns, variance and covariance for the next years:

$$\mu = \begin{pmatrix} 0.10 \\ 0.14 \\ 0.20 \end{pmatrix}, \quad \Sigma = \frac{1}{100} \begin{pmatrix} 0.2500 & 0.1050 & 0.1500 \\ 0.1050 & 0.4900 & 0.2100 \\ 0.1500 & 0.2100 & 1.000 \end{pmatrix}$$

Furthermore, it can be stated that

$$\Sigma^{-1} = \begin{pmatrix} 464.2857 & -76.5306 & -53.5714 \\ -76.5306 & 236.8805 & -38.2653 \\ -53.5714 & -38.2653 & 116.0714 \end{pmatrix}$$

1. Compute the correlation matrix.
2. Your required expected return is assumed to be $\mu_p = 0.15$. Which portfolio of the three assets should you choose? What is the risk of the portfolio?
3. Calculate global MVP, as well as its expected return and risk. Compare your result with a portfolio, only containing the least risky asset A.
4. Assume the risk-free rate is $r_f = 0.07$. Determine the portfolio weights in the tangency portfolio as well as its expected return and risk.
5. What Sharpe ratio is traded in the market?

Exercise 4

A financial market is described by a one-period state model. Two assets exist in this economy, and both assets cost 13.5 at time 0. At time 1 there are three possible states, and the assets pay a state dependent dividend, described by a vector d^i with entrance d_s^i , where i is the asset and s is the state, $i = 1, 2$ and $s = 1, 2, 3$. The economy can be described as:

$$P = \begin{pmatrix} P_1 \\ P_2 \end{pmatrix} = \begin{pmatrix} 13.5 \\ 13.5 \end{pmatrix}, \quad D = \begin{pmatrix} d_1^1 & d_2^1 & d_3^1 \\ d_1^2 & d_2^2 & d_3^2 \end{pmatrix} = \begin{pmatrix} 10 & 15 & 20 \\ 20 & 15 & 10 \end{pmatrix}$$

Furthermore, it can be stated that

$$P_1 = 0.25d_1^1 + 0.40d_2^1 + 0.25d_3^1 = 0.40d_1^1 + 0.10d_2^1 + 0.40d_3^1$$

$$P_2 = 0.25d_1^2 + 0.40d_2^2 + 0.25d_3^2 = 0.40d_1^1 + 0.10d_2^1 + 0.40d_3^1$$

1. Decide if the market is arbitrage free and complete.
2. Determine the amount of possible state prices in this market (Hint: These must fulfill that $P = D\psi$ is consistent and $\psi \gg 0$)

A new asset is added to the market. The price of the asset is $P_3 = 14$ and have dividend $d_1^3 = 20$, $d_2^3 = 10$ and $d_3^3 = 20$ in the three different states.

3. Decide if the expanded model is arbitrage free and complete. If yes, determine the state prices.
4. Calculate the risk-free rate and the risk neutral probabilities.
5. Consider the derivative asset, which payoff function is given by:

$$d_s^{derivative} = \max(d_s^2 d_s^3 - 250.0) \text{ for } s = 1, 2, 3.$$

Calculate the arbitrage free price of the asset.

Investment and Finance

Re-exam August, 2020

Assignment 1 (25%)

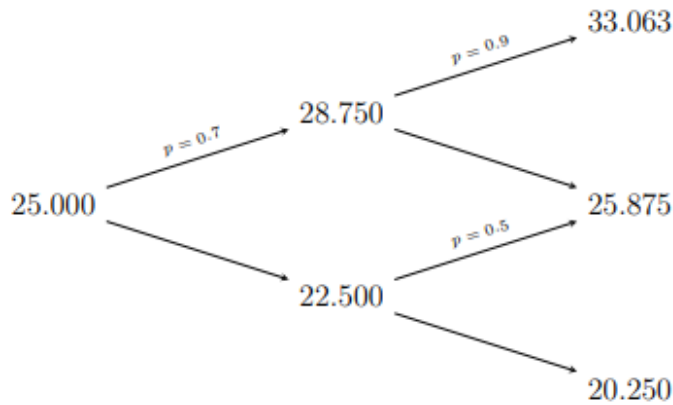
In a market of bonds, a variety of bonds are traded. The following table presents the most crucial information regarding the bonds, all of them have a principal of 100 DKK and yearly fixed terms:

Obligation	Maturity	Coupon	Price
Coupon bond	1 year	5%	98.700
Annuity bond	3 years	6%	97.269
Serial bond (fixed principal payments per year)	2 years	7%	100.125

1. Write the cash flows of the three bonds.
2. Calculate the discount factors in the market. Explain whether the market is free of arbitrage and complete.
3. What is the one-year spot rate in the market? What quarterly compounded interest rate is equivalent to the one-year spot rate?
4. What is the 3-year swap rate?
5. Consider a one-year forward contract, where the annuity bond is traded for take-over right after the term payment. What is the arbitrage-free forward price?
6. A firm has a debt and should at times 1, 2, and 3 pay 300, 250 and 200 DKK, respectively. How many of the bonds should the firm buy at time 0 in order to meet its debt obligations?

Assignment 2 (25%)

A share costs DKK 25 at time 0 in a market where the annually compounded risk-free rate is constant 7%. During the following two years, it is assumed that the price of the share may rise or fall according to the following binomial grid, where p , as usual, denotes the probability that the share will rise in that year:



1. Determine if the model is arbitrage-free and complete.
2. Calculate the risk-neutral probabilities, q and $1 - q$.
3. Find the futures price for a futures contract that expires at time 2.
4. What payments does the futures contract give rise to at times 1 and 2?
5. What should a European put option with strike 27.5 and expiration at time 2 cost in this market?
6. How can the put option be replicated with the risk-free asset and futures contracts at time 0?

Now assume that the share at time 1 pays a proportional dividend of 10% of the current price. All other prerequisites are unchanged.

7. Calculate the new price of the European put option.

Assignment 3 (25%)

As always the goal is to obtain good returns. Based on the recent movements of the financial markets, you identified three risky assets that you are interested in investing in for the "long term" portfolio. It is assumed that the rates of return on these assets follow a multi-normal distribution $r \sim N(\mu, \Sigma)$ where $r = [r_1, r_2, r_3]'$, r_i denote the return of the i 'th asset. Information about the three assets are given:

$$\mu = \begin{bmatrix} 0.08 \\ 0.12 \\ 0.10 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} 0.0144 & 0.0054 & 0.0078 \\ 0.0054 & 0.0225 & 0.0078 \\ 0.0078 & 0.0078 & 0.0169 \end{bmatrix}$$

1. What is the return and risk of an equally weighted portfolio of the three assets?

2. What is the probability that you obtain a negative return, when investing in the equally weighted portfolio?
3. If your required expected return is 0.15, which portfolio should you invest in? What is the associated risk?
4. Assume that the risk-free interest rate is $r = 0.03$. Calculate the weights of the tangency portfolio.
5. Write down the capital market line and depict it in a (σ, μ) -diagram. Plot the three assets in the diagram. For each asset, indicate graphically what proportion of the total risk is respectively systematic and unsystematic.
6. Calculate the β 's of each of the three assets.

Assignment 4 (25%)

A factor model is defined with the following parameters:

$$\begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix} = \begin{bmatrix} 0.10 \\ 0.15 \\ 0.20 \end{bmatrix} + \begin{bmatrix} 1.0 & 0.5 \\ 1.5 & 1.0 \\ 1.0 & 2.0 \end{bmatrix} \begin{bmatrix} F_1 \\ F_2 \end{bmatrix}$$

Here F_1 and F_2 are independent factors with mean 0 and variance 0.01.

1. Calculate the expected value and the standard deviation of the three rates of return.
2. Calculate the correlation between the rates of return.
3. Find the weights of the portfolio that yield a risk-free return. What is the risk-free return in this economy?
4. Use the three assets 1, 2 and 3 to construct pure factor-portfolios P_1 and P_2 . This means that you should find the three weights $(\omega_1, \omega_2, \omega_3)$, so the associated portfolio satisfies the factor equation:

$$r_{P_1} = \mu_{P_1} + F_1,$$

$$r_{P_2} = \mu_{P_2} + F_2.$$

5. What is the risk-premiums, λ_1 and λ_2 , in this model?